

Carl Edward Fields, Jr.

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Steward Observatory & Department of Astronomy, 933 N Cherry Ave, N504, Tucson, AZ 85719, USA

RESEARCH INTERESTS

Stellar Hydrodynamics and Nucleosynthesis, Core-Collapse Supenovae, Gravitational Waves, and Astronomy Education.

RECENT APPOINTMENTS

• Steward Observatory & Department of Astronomy, University of Arizona <i>Assistant Professor & Assistant Astronomer</i>	Aug. 2024 - Tucson, AZ
• Computational Physics and Methods Division, Los Alamos National Laboratory <i>Feynman Distinguished Postdoctoral Fellow</i>	May 2021 - May 2024 Los Alamos, NM
• LSST Corporation <i>Data Science Fellow</i>	Aug. 2017 - May 2019 East Lansing, MI
• Department of Physics & Astronomy, Michigan State University <i>NSF Graduate Research Fellow</i>	May 2016 - May 2021 East Lansing, MI
• Department of Physics & Astronomy, Michigan State University <i>FORD Foundation Predoctoral Fellow</i>	May 2016 - May 2021 East Lansing, MI

EDUCATION

• Michigan State University <i>Ph.D., Astronomy & Astrophysics - Multidimensional Hydrodynamic Simulations of Massive Stars and Their Explosions</i>	May 2021 East Lansing, MI
• Michigan State University <i>Certificate In College Teaching - College of Natural Sciences</i>	May 2020 East Lansing, MI
• Michigan State University <i>Certificate in High Performance Computing - Department of Computational Mathematics, Science and Engineering</i>	December 2018 East Lansing, MI
• Arizona State University <i>B.S., Physics, w/ Honors - College of Liberal Arts and Sciences</i>	May 2016 Tempe, AZ
• Arizona State University <i>B.S., Astrophysics, w/ Honors - School of Earth & Space Exploration</i>	May 2016 Tempe, AZ

RECENT FUNDING, AWARDS, DISTINCTIONS

• Scialog Fellow, Research Corporation For Science Advancement <i>Early Science with the LSST</i>	Nov. 2024 -
• Director's Discretionary Computing Allocation (PI), Argonne National Laboratory <i>5,000 Node-Hours on polaris</i>	2024 -
• Benjamin Dean Astronomy Lecture, Morrison Planetarium, California Academy of Sciences <i>The Remarkable Deaths of Massive Stars</i>	Dec. 2023
• SPOT Award, Los Alamos National Laboratory <i>for outstanding performance and lasting contributions to . . .</i>	2022
• Institutional Computing Award (PI), Los Alamos National Laboratory <i>12,000,000 SU</i>	2021 - 2023
• FORBES 30 under 30, Science, FORBES <i>Featured Honoree</i>	2021
• Dr. Pliny A. and Margaret H. Price Prize, CCAPP, Ohio State University <i>research excellence and exceptional promise</i>	2020
• Edward A. Bouchet Graduate Honor Society, Michigan State University <i>exemplify the five core values of the society</i>	2020
• Beth Brown Memorial Prize, American Astronomical Society/National Society of Black Physicists <i>best graduate oral presentation</i>	2020

RECENT PUBLICATIONS

1. *AMReX-Astrophysics Microphysics: microphysics routines for astrophysical simulation codes based on the AMReX library*, The AMReX-Astrophysics Microphysics Team including **C. E. Fields**, *Journal of Open Science*, under review, 2025.
2. *Phoebus: Performance Portable GRRMHD for Relativistic Astrophysics*
B. Barker et al. including **C. E. Fields**, *Astrophysical Journal*, under review, (2025).
3. *Enhanced production of ⁶⁰Fe in massive stars*
A. Spyrou et al. including **C. E. Fields**, *Nature Communications* Vol. 15, Article number: 9608 (2024).
4. *The Three-Dimensional Collapse of a Rapidly Rotating 16 M_⊙ Star*
C. E. Fields, 2022 *ApJL*, **924**, L15

ADVISING AND MENTORING

- **Logan White** August 2025 -
University of Arizona, Department of Astronomy, PhD Advisor
- **Maria Mutz** January 2025 -
University of Arizona, Department of Physics, PhD Advisor
 - NSF ACCESS Computing Award: (2025-2026), 1.5M ACCESS Credits, *Stampede 3*, Texas Advanced Computing Center
- **Camille Ciofi** August 2024 -
University of Arizona, Undergraduate Research Advisor
 - TIMESTEP Research Apprenticeship (2024-2025)
 - NASA Space Grant Undergraduate Research Internship Program (2025-2026)
- **Jennifer Stafford** June 2024 -
University of Wisconsin (Madison), Graduate Research Mentor

TEACHING AND EDUCATION

- **Theoretical Astrophysics I: Stellar, AST400A** Steward Observatory, Fall 2025
Undergraduate Astrophysics Major Course [Course Materials 
- **Lamat Institute - MESA+FLASH Mini-Workshop** [Workshop Materials 
University of California, Santa Cruz, August 2025
- **Astrophysics of Stars and Planets, AST/PTYS545** Steward Observatory, Spring 2025
Graduate Course [Course Materials 
- **LSSTC Data Science Fellowship Program - Session 15** CfA, Harvard University | July 2022, Cambridge, MA
Guest Lecturer, Graphics Processing Units & Parallel Computing in Python [Course Materials 
- **Stellar Modeling for Nuclear Astrophysics Conference** Remote, 2022
Lecturer, Custom nuclear reaction rates in MESA [Course Materials 
- **Visualizing Scientific Datasets** Michigan State University, Spring 2022
Graduate Teaching Assistant, Department of Computational Mathematics, Science and Engineering

RECENT NATIONAL AND INTERNATIONAL SERVICE

- **Executive Committee** 2025-2027
Center for Nuclear Astrophysics Across Messengers, Facility for Rare Isotope Beams
- **Early Career Member-at-Large, Executive Committee** 2022-2024
American Physical Society, Division of Astrophysics
- **Scientific Reviewer/Panelist** 2021-
NSF, NASA, DOE
- **Local Organizing Committee - JINA Frontiers 2019 Meeting** 2019
Joint Institute for Nuclear Astrophysics, MSU
- **Scientific Referee** 2016-
AAS Journals, A&A

RECENT UNIVERSITY AND DEPARTMENT SERVICE

- **Theoretical Astrophysics Program (TAP) Steering Committee** 2024-
Steward Observatory, Department of Astronomy, University of Arizona
- **SO/DA Advisory Committee to the Director** 2024-
Steward Observatory, Department of Astronomy, University of Arizona
- **Graduate Student First Year Committee** 2024-2025
Steward Observatory, Department of Astronomy, University of Arizona
- **Theoretical Astrophysics Program (TAP) Subcommittee (Graduate Research Prize)** 2024
Steward Observatory, Department of Astronomy, University of Arizona
- **Bok Fellowship Search Committee** 2024
Steward Observatory, Department of Astronomy, University of Arizona
- **Graduate Admissions Committee** 2024
Steward Observatory, Department of Astronomy, University of Arizona

RECENT PUBLIC ENGAGEMENT AND OUTREACH

- **Public Evening Lecture** Oct. 2025
Steward Observatory, University of Arizona 
- **Benjamin Dean Public Lecture** Dec. 2023
Morrison Planetarium, California Academy of Sciences 
- **Speaker's Night, a public lecture** April 2022
RASC, Toronto Centre 
- **SpaceTime Expert** 2018-2020
SpaceTime Program, The Planetary Society 

RECENT PRESENTATIONS (INVITED)

- **Ohio State University, Center for Cosmology and AstroParticle Physics**
Astronomy Colloquium | March 2025
- **University of California, Santa Cruz**
Astronomy Colloquium | Nov 2024
- **Michigan State University**
JINA-CEE Frontiers Junior Workshop Talk + Panel | May 2023
- **University of Michigan**
Astronomy Colloquium | April 2023
- **Center for Astrophysics | Harvard & Smithsonian**
Institute for Theory and Computation Colloquium | April 2023
- **Carnegie Observatories**
Astronomy Colloquium | March 2022
- **Stockholm University, The Oskar Klein Centre**
Astronomy Colloquium | February 2022
- **University of California (San Diego), Department of Physics**
Astronomy Seminar | January 2022
- **Kavli Institute for Theoretical Physics, Probes of Transport in Stellar Interiors**
Invited Talk | November 2021
- **University of Texas, Austin, Department of Astronomy**
Colloquium | October 2021
- **NASA, Goddard Space Flight Center**
ASD Colloquium | April 2021
- **Harvard University, Center for Astrophysics**
GCSP Seminar | November 2020
- **California Institute for Technology, Department of Astronomy**
Astronomy Tea Talk | November 2020
- **University of Massachusetts (Dartmouth), Department of Physics**
Physics Colloquium | October 2020
- **University of California (Berkeley), Department of Astronomy**
Astro Lunch Seminar | October 2020
- **University of Connecticut, Department of Physics**
Astronomy Seminar | October 2020
- **Ohio State University, Center for Cosmology and AstroParticle Physics**
Price Prize Seminar | September 2020
- **Stanford University, Kavli Institute for Particle Astrophysics and Cosmology**
Tea Talk | September 2020
- **Flatiron Institute, Center for Computational Astrophysics**
Compact Object Group Meeting Talk | September 2020
- **Harvard University, The Banneker Institute**
Oral Presentation | July 2019 - Cambridge, Massachusetts
- **University of Tokyo, Kavli Institute for Physics & Mathematics of The Universe**
Seminar | January 2019 - Kashiwa, Japan

RECENT PRESENTATIONS (CONTRIBUTED)

- **Conference of The National Society of Black Physicist**
Oral Presentation | November 2019 - Providence, Rhode Island
- **Los Alamos National Laboratory, Center for Theoretical Astrophysics**
Astro Seminar | July 2018 - Los Alamos, NM, USA
- **231st Meeting of The American Astronomical Society**
Student Presenter | January 2018 - Washington, DC
- **Conference of Ford Fellows**
Student Presenter | (Postponed May 2018) - San Juan, PR
- **JINA-CEE Frontiers in Nuclear Astrophysics**
Student Presenter | February 2017 - East Lansing, MI
- **National Society of Black Physicists Fall Workshop**
Student Presenter | October 2016 - Fermilab
- **AGEP Michigan Alliance 2016 Fall Conference**
Student Presenter | October 2016 - East Lansing, MI
- **Conference of Ford Fellows**
Student Presenter | September 2016 - East Lansing, MI
- **227th Meeting of The American Astronomical Society**
Student Presenter | January 2016 - Kissimmee, FL

AWARDS, HONORS, DISTINCTIONS, FUNDING

- **Scialog Fellow**, Research Corporation For Science Advancement Nov. 2024 -
Early Science with the LSST
- **Director's Discretionary Computing Allocation (PI)**, Argonne National Laboratory 2024 -
5,000 Node-Hours on [polaris](#)
- **Benjamin Dean Astronomy Lecture**, Morrison Planetarium, California Academy of Sciences Dec. 2023
The Remarkable Deaths of Massive Stars
- **SPOT Award**, Los Alamos National Laboratory 2022
for outstanding performance and lasting contributions to . . .
- **Institutional Computing Award (PI)**, Los Alamos National Laboratory 2021 - 2023
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- **RPF Distinguished Postdoctoral Fellow**, Los Alamos National Laboratory 2021
extraordinary ability in scientific research . . .
- **Hubble Fellowship**, NASA (declined) 2021
exceptionally promising and innovative early-career astrophysicists
- **Forbes 30 Under 30**, [Science](#) 2021
Featured Honoree
- **Dr. Pliny A. and Margaret H. Price Prize**, CCAPP, Ohio State University 2020
research excellence and exceptional promise
- **Edward A. Bouchet Graduate Honor Society**, Michigan State University 2020
exemplify the five core values of the society
- **Beth Brown Memorial Prize**, American Astronomical Society/National Society of Black Physicists 2020
best graduate oral presentation
- **Data Science Fellowship**, LSST Corporation 2017 - 2019
positions graduate students to meet the scientific challenges of large astronomy datasets
- **Graduate Research Fellow**, NSF 2016 - 2021
recognizes and supports outstanding graduate students . . .
- **Predoctoral Fellow**, FORD Foundation 2016 - 2021
prepare them for the pursuit of a career in academic teaching or research. . . .
- **Chambliss Astronomy Achievement Award (Hon. Ment.)**, American Astronomical Society 2016
undergraduate poster presentation
- **Space Grant Undergraduate Fellowship**, NASA 2015 - 2016
academic achievement, research experience
- **Carl A. Rouse Fellowship**, National Society of Black Physicists/Caltech 2016
in support of research in gravitational-wave science
- **LIGO Summer Undergraduate Research Fellowship**, Caltech 2015
research experience, academic achievements
- **Beth Brown Memorial Prize**, National Society of Black Physicists/American Astronomical Society 2015
best undergraduate poster presentation
- **Norm Perrill Origins Project Undergraduate Scholar**, Arizona State University Origins Project 2015
academic achievement, origins based research

SELECTED PRESS/INTERVIEWS

- **Featured Scientist, University Physics with Modern Physics, 16th Edition (Textbook)** 2025
Pearson Publishing
- **Benjamin Dean Astronomy Lecture and Public Interview** 2023
Morrison Planetarium, California Academy of Sciences
- **MSU astrophysics doctoral student Carl Fields named one of Forbes' 30 Under 30 for science** 2020
Lansing State Journal, Newspaper
- **Black voices in physics: Carl Fields** 2020
Physics Today, Interview, Journal
- **2020 Price Prize Winners Announced** 2020
Center for Cosmology and AstroParticle Physics, OSU
- **Study of the Evolution of Stars and Stellar Explosions** 2020
Institute for Cyber Enabled Research, MSU, Interview
- **Determination is the key** 2016
ASU Now, Interview

REFEREED PUBLICATIONS (AS 1st AUTHOR)

7. *A new stellar model for massive stars*
C. E. Fields, C. Cioffi, S. M. Couch, and J. Goldberg, 2026, *ApJ*, in prep.
6. *MESA-Web: A cloud resource for stellar evolution in astronomy curriculum*
C. E. Fields et al. (2023), *AEJ*, [Vol. 3 No. 1: Astronomy Education Journal](#)- December 2023.
5. *The Three-Dimensional Collapse of a Rapidly Rotating 16 M_⊙ Star*
C. E. Fields, 2022 *ApJL*, **924**, L15
4. *Three-dimensional Hydrodynamic Simulations of Convective Nuclear Burning in Massive Stars Near Iron Core Collapse*
C. E. Fields & S. M. Couch, 2021, *ApJ*, 2021 *ApJ*, **921**, 28
3. *On The Development of Multidimensional Progenitor Models For Core-collapse Supernovae*
C. E. Fields & S. M. Couch, 2020, *ApJ*, 2020 *ApJ*, **901**, 33
2. *The Impact of Nuclear Reaction Rate Uncertainties On The Evolution of Core-Collapse Supernova Progenitors*
C. E. Fields, R. Farmer, I. Petermann, S. M. Couch, & F. X. Timmes, 2018 *ApJS*, **867**, 30
1. *Properties of Carbon-Oxygen White Dwarfs From Monte Carlo Stellar Models*
C. E. Fields, R. Farmer, I. Petermann, C. Iliadis, & F. X. Timmes, *ApJ*, **823**, **46**, 2016

REFEREED PUBLICATIONS (AS CO-AUTHOR AUTHOR)

12. *Multi-dimensional MHD Simulations of Turbulent Convection in Massive Stars*,
M. Mutz & **C. E. Fields**, 2026, *ApJ*, in prep.
11. *AMReX-Astrophysics Microphysics: microphysics routines for astrophysical simulation codes based on the AMReX library*,
The AMReX-Astrophysics Microphysics Team including **C. E. Fields**, *Journal of Open Science*, under review, 2025.
10. *Phoebus: Performance Portable GRRMHD for Relativistic Astrophysics*
B. Barker et al. including **C. E. Fields**, *Astrophysical Journal*, under review, (2025).
9. *Enhanced production of ⁶⁰Fe in massive stars*
A. Spyrou et al. including **C. E. Fields**, *Nature Communications* Vol. 15, Article number: 9608 (2024).
8. *Testing White Dwarf Age Estimates using Wide Double White Dwarf Binaries from Gaia EDR3*
Heintz, Tyler and 87 other co-authors including **C. E. Fields**, *ApJ*, **934**, 148 (2022).
7. *On the Impact of ²²Ne on the Pulsation Periods of Carbon-Oxygen White Dwarfs with Helium-dominated Atmospheres*
Chidester, Morgan and 8 other co-authors including **C. E. Fields**, *ApJ*, **910**, 24 (2021).
6. *Evolutionary roads leading to low effective spins, high black hole masses, and O1/O2 rates for LIGO/Virgo binary black holes*
K. Belczynski, J. Klencki, & **C. E. Fields** et al., *A&A*, **636**, A104 (2020).
5. *Understanding the Engines and Progenitors of Gamma-Ray Bursts*
C. L. Fryer and 7 other co-authors including **C. E. Fields**, *Eur. Phys. J. A* (2019) **55**: 132.
4. *The Impact of White Dwarf Luminosity Profiles on Oscillation Frequencies*
F. X. Timmes, Richard H. D. Townsend, Evan B. Bauer, Anne Thoul, **C. E. Fields** & William M. Wolf, 2018 *ApJL*, **234**, 19.
3. *The ¹²C(α, γ)¹⁶O Reaction and Its Implications for Stellar Helium Burning*
R. J. deBoer and 12 other co-authors including **C. E. Fields**, *Rev. Mod. Phys.* **89**, 035007 (2017).
2. *On Variations Of Pre-Supernova Model Properties*
R. Farmer, **C. E. Fields**, I. Petermann, L. Dessart, M. Cantiello, B. Paxton, & F. X. Timmes, *ApJS*, 2016, **227**, 22.
1. *On Carbon Burning in Super Asymptotic Giant Branch Stars*
R. Farmer, **C. E. Fields**, & F. X. Timmes, *ApJ*, **807**, **184**, 2015.

PUBLICATIONS SUMMARY

Citations: 1491; *h*-index: 15; *i*₁₀-index: 15; *i*₁₀₀-index: 3 ([Google Scholar](#)) | 21 publications: 18 refereed, 2 submitted, 2 in prep.